

M.V.P Samaj's,

K. R. T. ARTS, B. H. COMMERSE, A. M. SCIENCE (K.T.H.M.) COLLEGE, NASHIK-02.

PROJECT REPORT ON

"STATISTICAL ANALYSIS ON FAST-FOOD CONSUMPTION & IT'S IMPACT ON HEALTH"

UNDER THE GUIDENCE OF

MS. SNEHAL A. KAWADE

DEPARTMENT OF STATISTICS

K.T.H.M. COLLEGE NASHIK.

REPORTED BY

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K.T.H.M. COLLEGE, NASHIK.

(Re-accredited 'A++' by NAAC)

AFFILIATED TO

SAVITRIBAI PHULE PUNE UNIVERSITY, PUNE. ACADEMIC YEAR 2021-2022.

K.R.T. Arts, B.H. Commerce, A.M. Science College, Nashik.

("A++" Grade Re-accredited by NAAC)

CERTIFICATE

DEPARTMENT OF STATISTICS

This is to certify, that

1] Koshti Savita Rajendra

2] Mahale Shubhangi Rajendra

3] Nagare Ishwari Sanjay

4] Patil Bhavesh Dilip

5] Somwanshi Pawan Kailas

of T. Y. B.Sc. (STATISTICS) has completed the project, under guidance of Prof. Ms. Snehal Kawade during the year 2021-2022, at the department of statistics in K.T.H.M. College Nashik, affiliated to Savitribai Phule Pune University, Pune, Maharashtra.

This project work was carried out for partial fulfilment of the **Degree of Bachelors Of Science In Statistics**.

Ms. Snehal A. Kawade	**Examiner**	**Dr. G. S. Phad**
Guide		Head of Department Of
Department of Statistics		Statistics

Date: -

Place: - Nashik

ACKNOWLEDGEMENT

An accomplishment and final outcome of this project required a lot of guidance and assistance from many people and I am extremely privileged to have got this all along the completion to my project. All that I have done only due to such supervision and assistance and I would not forget to thank them.

I respect and thank Dr. Ganesh S. Phad, Head Department of Statistics and also an experienced project guide, for providing me opportunity to do the project work and giving us all support and guidance, which made me complete the project duty, I am extremely thankful to him providing such a nice support and guidance, although she had busy schedule managing the regular lectures as well as important meetings.

I own my deep gratitude to our project guide Ms. Snehal A. Kawade, Assistant professor, Department of Statistics, who took interest on our work and guided us all along, till the completion of our project work by providing all the necessary information for developing a good system.

I am thankful to and fortunate enough to get constant encouragement, support and guidance from all teaching staff, Department of Statistics who helped us in successfully completing our project work.

- Koshti Savita Rajendra.
- Mahale Shubhangi Rajendra.
- Nagare Ishwari Sanjay.
- Patil Bhavesh Dilip.
- Somwanshi Pawan Kailas.

DECLARATION

We declare that the project entitled “Statistical Analysis on Fast-food Consumption & It’s Impact on Health, submitted by us, for the partial fulfilment of our bachelor degree of science in statistics (2021-22) is our original work.

We further declare that the analysis has been carried out based on the primary data collected from google form survey. We have given our best and hope that our project work may be helpful to improve awareness about eating lifestyle.

- Koshti Savita Rajendra.
- Mahale Shubhangi Rajendra.
- Nagare Ishwari Sanjay.
- Patil Bhavesh Dilip.
- Somwanshi Pawan Kailas.

Date: -

Place: - Nashik.

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INTRODUCTION

Concept, Relationship, lifestyles are metamorphosed to accommodate the new jet age and feeding habits too is no exception. Health wholesome food have been replaced by the new food mantra-JUNKFOOD! Within the context of world economy. junk food could be a global phenomenon. The availability of food and snacks at low prices and selling methods tailored by manufacturers of such food has triggered an evolution whereby, consumption of food that require neither the structure nor the preparation of a proper meal. It appears to have engulfed each age; each race and also the newest entrants on stage are children, faculty going in particular. Hence, a scientific presentation has been created during this review from the article from various source highlighting feeding habits, nutritional aspects and quality of unhealthy food, their health impact on consumption and preventive measures to be undertaken.




KFC


McDonald's
DRIVE THRU


Pizza Hut


Domino's Piz
GREAT
JOB

ABSTRACT

Eat healthy and live healthy is one in all the essential necessities for long time. Sadly, today's world has been adapted to a system of consumption of food that has many adverse effects on health. way changes have compelled us most that one has thus very little time to actually suppose what we tend to area unit consumption is right! Economic process and urbanisation have greatly affected one's consumption habit and compelled many folks to consume fancy and high calorie quick food, popularly called "junk food". The aim of the study is fast food consumption pattern and health problems. The method and materials used in the doctrinal method. The result of the paper fast food culture among teenagers has been increased. The conclusion of the paper is junk food have actually sliced up the Third World due to economic process. It's an integral a part of life within the developed and also the developing world**.**

KEYWORDS

Fast food, consumption, youngsters, temptation.

OBJECTIVES

1] To Compare the trends of fast-food consumption according to age, gender, financial status.

2] To know the reasons behind the consumption of fast-food.

3] To find which is the most preferred fast-food.

4] To find out which is the most preferred place and time for the consumption of fast-food.

5] To get information about average amount of money that people spend on fast-food.

6] To check the awareness about health risk caused by the consumption of fast-food.

7] To study the increasing temptation of youngsters on fast-food.

TOOLS

1] PYTHON (PYCHARM).

2] R - SOFTWARE.

3] MS – OFFICE.

METHODOLOGY

❖STATISTICAL METHODS

In this chapter we have given all the methods which are used in statistical analysis. The Statistical Methods is a technique used to obtain analysis and prepare the numerical data. Statistical methods simplify unwisely masses of complex data by processes of classification, tabulation, summarization, possibilities from the nature of the data available. It deals with all aspects of data including data collection in terms of the design of surveys and experiments.

On the basis of collected data to draw different conclusion and inference, the following Statistical Procedures are adopted:

1. Graphical Representation.
2. Testing Of Hypothesis by applying the chi-square test (Independence Of Attributes).
3. Decision Tree.

❖GRAPHICAL REPRESENTATION

Graphical Representation can be thought of as a way to obtain a "first look" at a group of data. It does not provide a definitive interpretation of the data, but can lead to an intuitive "feel" for the data.

However, this "first look" can be deceiving because the human mind wants to classify everything it views based on something it already knows. This can lead to an erroneous first impression that can be hard for a researcher to dismiss, even as evidence mounts that it may be wrong.

Keep in mind that the purpose of graphical methods is simply to get a first look at the data without drawing conclusion. It can, however lead to hypothesis that guide further investigation.

❖TESTING OF HYPOTHESIS

✠ CHI-SQUARE TEST OF INDEPENDENCE OF ATTRIBUTES:

Test of statistical hypothesis called chi-square test. A test of this sort was originally proposed by Karl Pearson in 1900 and it provided one of the earlier methods of statistical inference.

It is used to test the independence of two attributes through contingency table. The data was tabulated in the form of r*c tables.

According to prof. R.A. Fisher the following condition must be satisfied for the validity of the chi-square test.

A] The sample observation should be independent and normally distributed.

B] Constraints imposed upon the observations, if any should be linear.

C] Expected or observed frequency should not be less than 5.

D] A total frequency should be reasonably large, say more than 50, it means that sample size is large enough.

HYPOTHESIS:

H_0: Two variables are not independent.

H_1: Two variables are independent.

TEST STATISTICS:

The chi-square test statistic is given by-

$$\sum_{i=1}^{r}\sum_{j=1}^{c}\frac{(o_{ij}-e_{ij})^2}{e_{ij}}$$

, Where, i =1, 2, …. r; j =1, 2, …. c.

Which follows chi square distribution under the null hypothesis with the degree of freedom (r-1) * (c-1).

Where, r = no of rows.

c = no. of columns.

o_{ij} = Frequency observed in $(i,j)^{th}$ cell.

e_{ij} = Expected frequency calculated in $(i,j)^{th}$ cell.

$e_{ij} = n_{i.} * n_{.j}$

$n_{i.}$= Total frequency of ith row.

$n_{.j}$=Total frequency of jth column.

n=Total no. of observation.

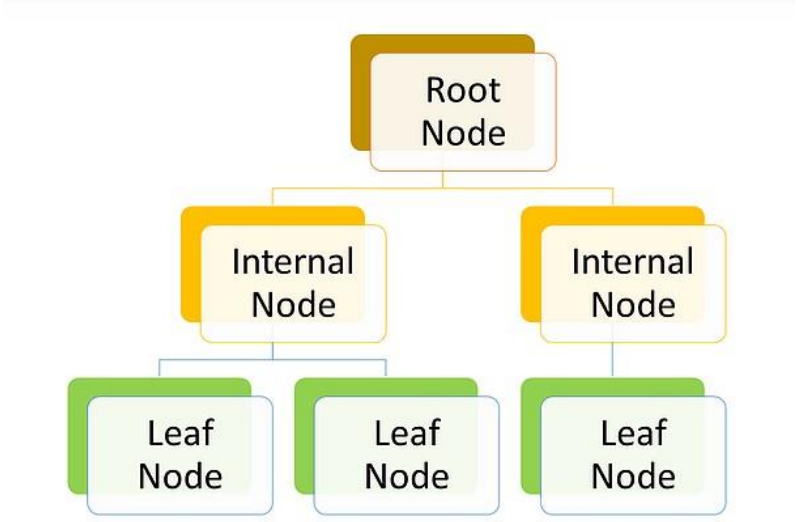
After calculating the value of chi-square statistics, we compare it with the tabulated value of chi-square statistics at 5% level of significance with (r-1) *(c-1) degree of freedom.

If the calculated value is less than tabulated value at 5% level of significance with (r-1) *(c-1)degree of freedom, we accept our null hypothesis and conclude that there is no association between two attributes otherwise we reject our null hypothesis and accept the alternative hypothesis, and conclude that is an association the two attributes.

❖DECISION TREE:

Decision tree is a supervised learning algorithm that works for both categorical and continuous input and output variables that is we can predict both categorical variables (classification tree) and a continuous variable (regression tree).

Decision tree is a flow chart like structure with the if-else condition. The top node is the root node, you start question from root node then move through the tree branches according to which groups you belong to until you reach a leaf node.



✠ Classification tree:

- A decision tree where target variable is categorical.
- The algorithm classifies the class within which the target variable would most likely to fall.
- It uses Gini index as metric/cost function to evaluate split in feature selection.
- Example like predicting who will or who will not order food, whether the weather will be rainy, sunny or cool.

DATA SUMMERIZATION, TABULATION AND GRAPHICAL REPRESENTATION

❖IN THE FOLLOWING MULTIPLE BAR DIAGRAMS ON CONSUMPTION OF FAST-FOOD ACCORDING TO AGE, GENDER, RESIDENTIAL AREA, RESIDENTIAL STATUS, OCCUPATION, FAMILY INCOME.

1]

	Below 10	10 – 25	25 -50	Above 50	Total
Yes	19	189	53	8	269
No	0	33	6	2	41

Table: Frequency of Consumption of Fast-food (According to Age)

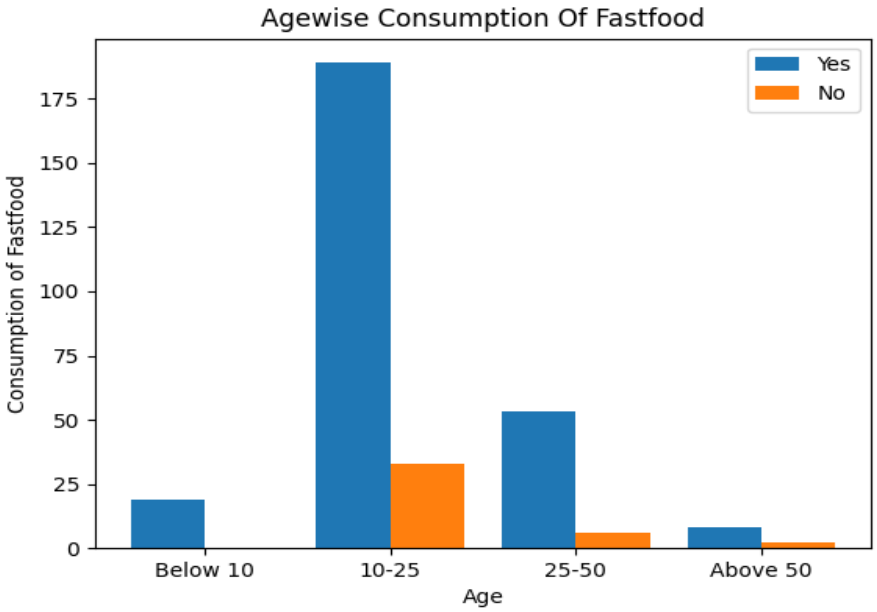


Figure: Graph of Fast-food Consumption (According to Age)

INTERPRETATION: In the above table there are 269 people who consuming fast-food out of which 19 people are below age 10, 189 belongs to the age group 10-25, 53 belongs to the age group 25-50 and 8 belongs to the group above 50. Whereas, there are 0,33,6,2 people of group below 10, 10-25, 25-50, Above 50 respectively not consuming fast-food.

2]

	Male	Female	Total
Yes	117	152	269
No	19	22	41

Table: Frequency of Consumption of Fast-food (According to Gender)

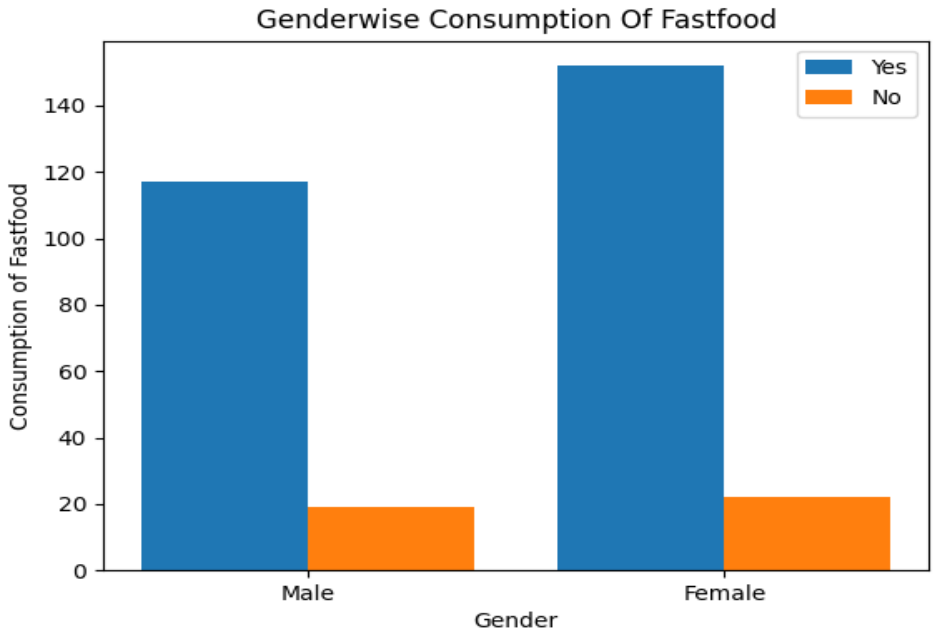


Figure: Graph of Consumption of Fast-food. (According to Gender).

INTERPRETATION: In the above table there are 269 Male & Female, out of which 117 Males and 152 Females consuming Fast-food. Whereas, there are 19 Male & 22 Female who are not consuming Fast-food.

3]

	Urban Area	Rural Area	Total
Yes	180	89	269
No	18	23	41

Table: Frequency of Consumption of Fast-food (According to Area)

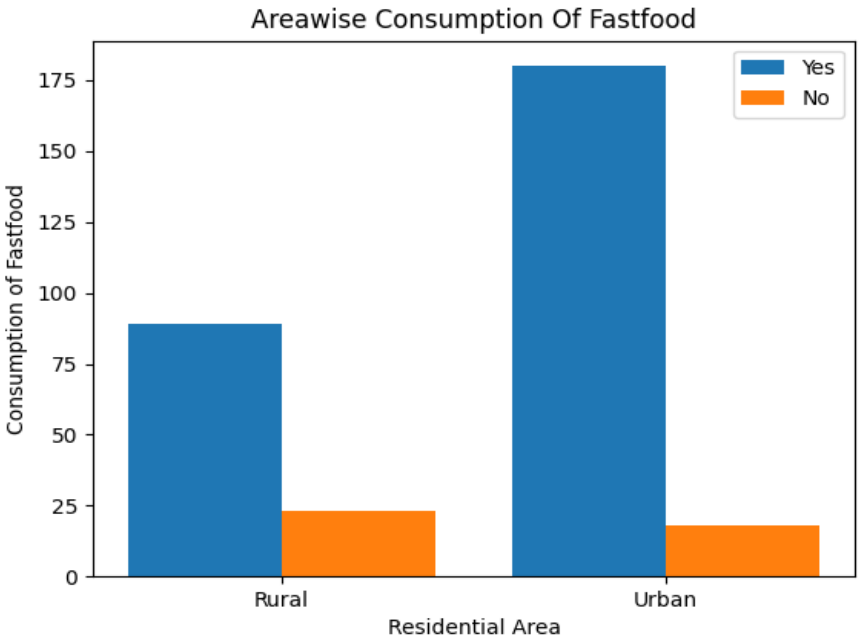


Figure: Graph of Consumption of Fast-food. (According to Gender).

INTERPRETATION: In the above table there are 269 People. out of which 89 people from Rural Area and 180 people from Urban Area consuming Fast-food. Whereas, there are 23 from rural & 18 from urban area who are not consuming Fast-food.

4]

	Yes	No
Students	197	36
Private Sector	35	3
Government Sector	10	0
Unemployed	5	0
Housewife	12	2
Business	6	0
Professor	4	0
Total	269	41

Table: Frequency of Consumption of Fast-food (According to Occupation)

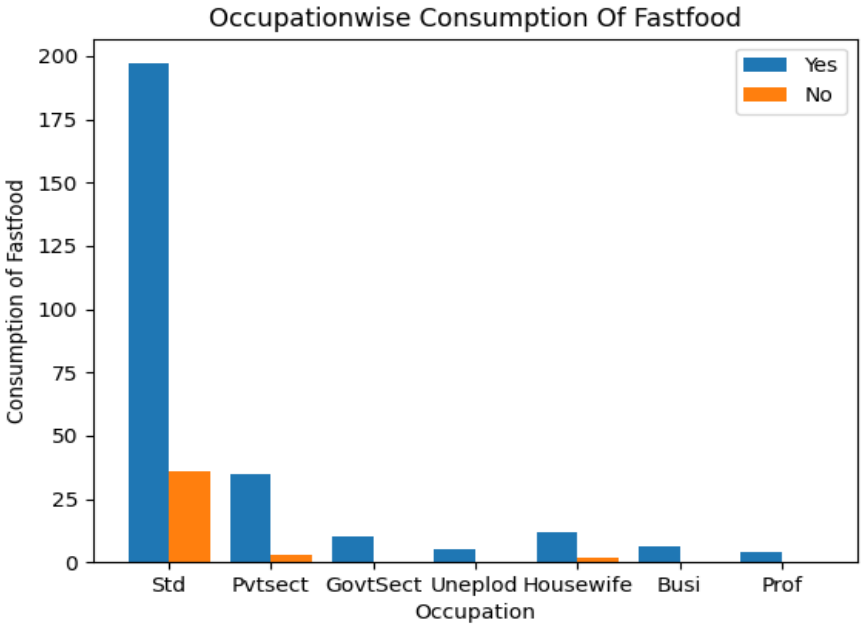


Figure: Graph of Consumption of Fast-food. (According to Occupation).

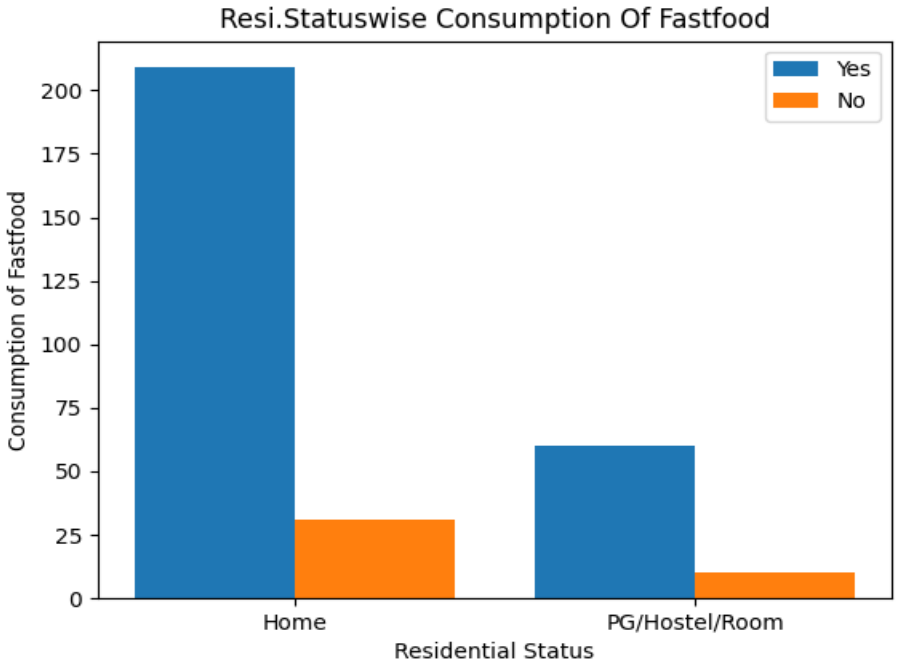
INTERPRETATION: In the above table there are 269 People who consuming fast-food. out of which 197 are Students, 35 are private sector employee, 10 are Government Sector Employee, 5 Unemployed, 12 Housewife's, 8 Businessmen, 4 are Professors. Whereas, there are Remaining 41 who are not consuming Fast-food.

5]

	Home	PG/Hostel/Rooms	Total
Yes	209	60	269
No	31	10	41

Table: Frequency of Consumption of Fast-food (According to Residential Status).

Figure: Graph of Consumption of Fast-food. (According To Residential Status).



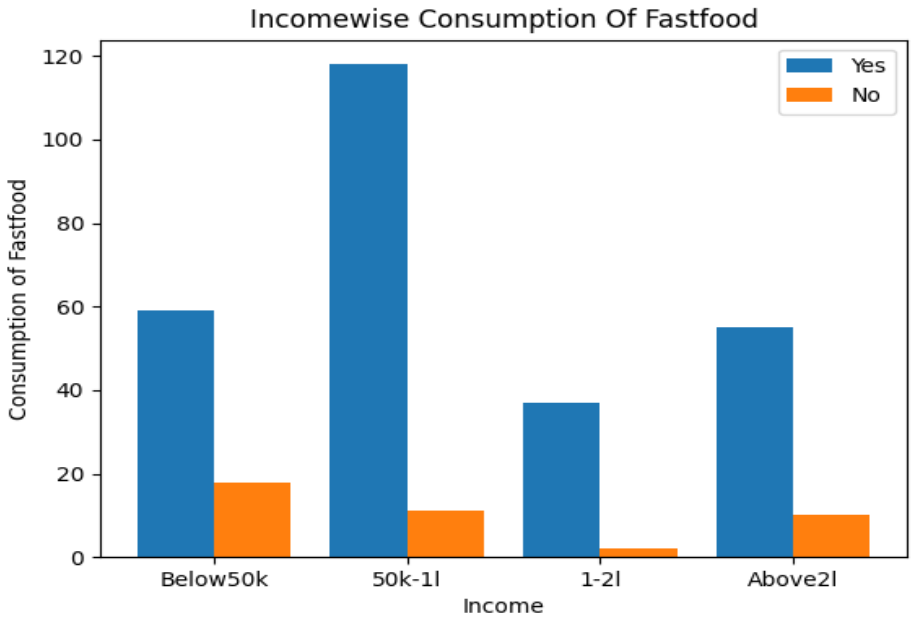
INTERPRETATION: In the above table there are 269 People who consuming fast-food. out of which 209 live at Home & 60 are at PG/Hostel/Rooms. Whereas, there are 31 & 10 live at Home & Hostel/PG/Rooms respectively who are not consuming Fast-food.

6]

	Below 50k	50k – 1 Lakh	1 – 2 Lakh	Above 2 lakhs	Total
Yes	59	118	37	55	269
No	18	11	2	10	41

Table: Frequency of Consumption of Fast-food. (According To Income).

Figure: Graph of Consumption of Fast-food. (According To Income).

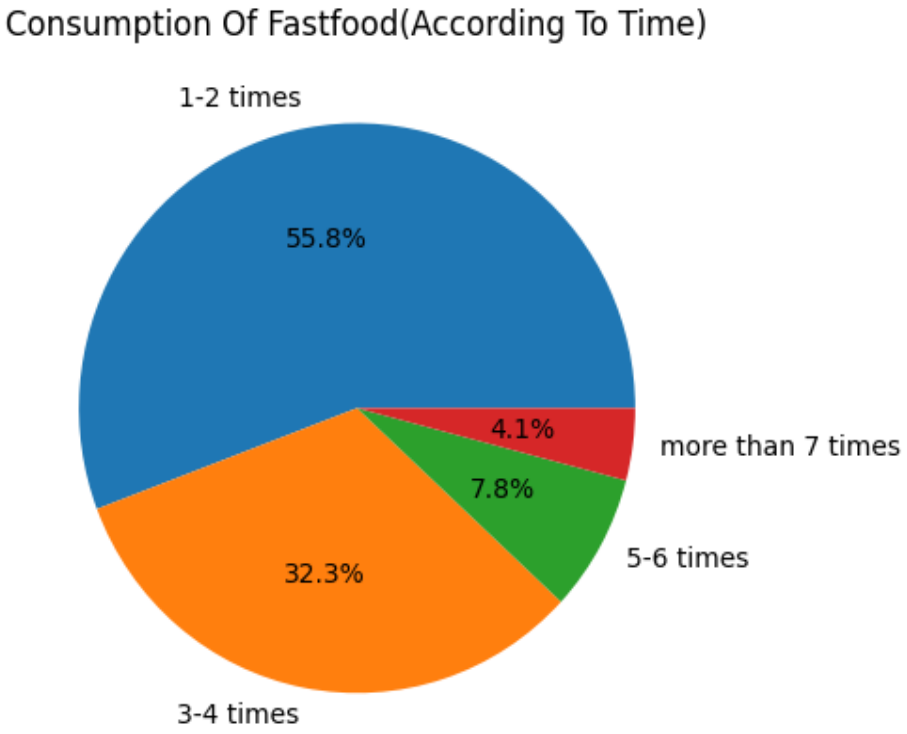


INTERPRETATION: In the above table there are 269 People who consuming fast-food. out of which 59 People has income Below 50000, 118 People has income 50000-1 lakh, 37 People has income 1 – 2 lakhs, 55 People has income Above 2 lakhs. Whereas 18,11,2,10 People has income Below 50000, 50000-1 lakh, 1-2 lakh, above 2 lakhs Are not consuming fast-food.

❖FREQUENCY TABLE AND THEIR CORRESPONDING PIE CHARTS FOR DIFFERENT REASONS THAT PEOPLE CONSUMED FAST-FOOD.

1]

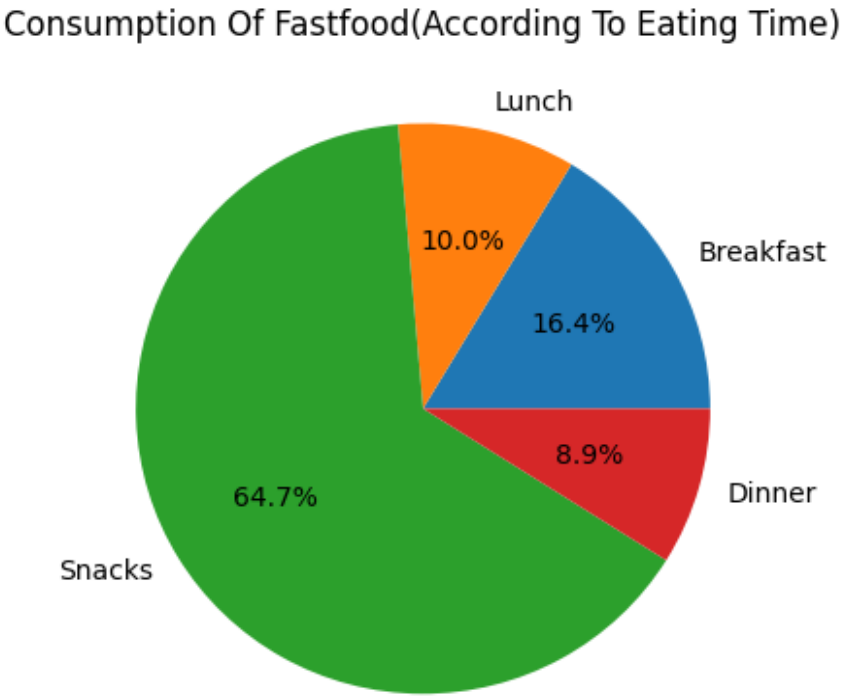
	FREQUENCY	PERCENTAGE
1-2 TIMES	150	55.8
3-4 TIMES	87	32.3
5-6 TIMES	21	7.8
MORE THAN 7 TIMES	11	4.1
TOTAL	269	100



INTERPRETATION: Proportion of people consuming fast-food 1-2 times (i.e.,55.8%) Is more than proportion of people consuming fast-food more than 7 times (i.e.,4.1%) (weekly basis).

2]

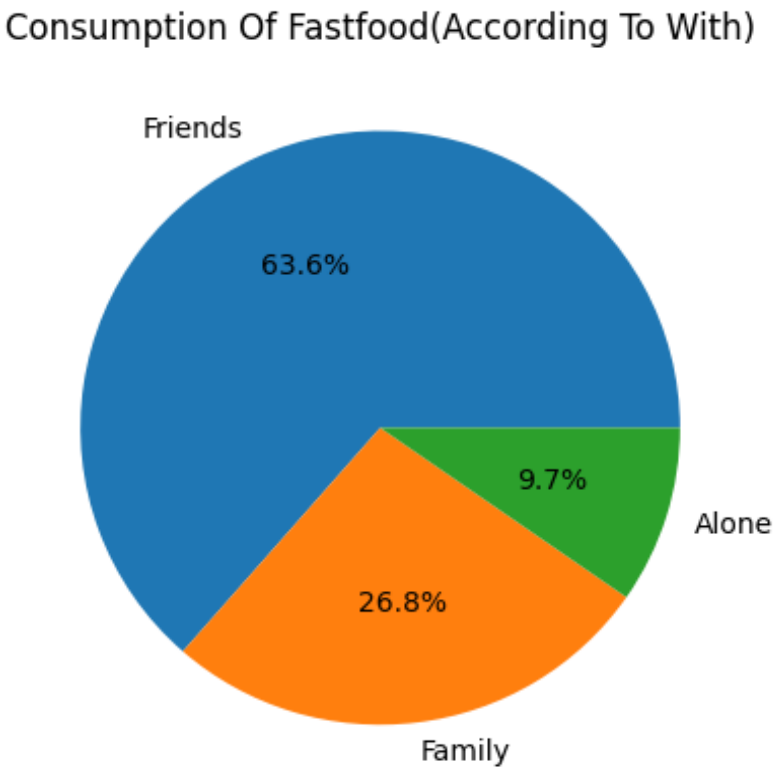
	FREQUENCY	PERCENTAGE
BREAKFAST	44	16.4
LUNCH	27	10.0
SNACK	174	64.7
DINNER	24	8.9
TOTAL	269	100



INTERPRETATION: Proportion of people consuming fast-food during snacks time (i.e.,64.7) is more than proportion of people consuming fast-food during dinner (i.e.,8.9%).

3]

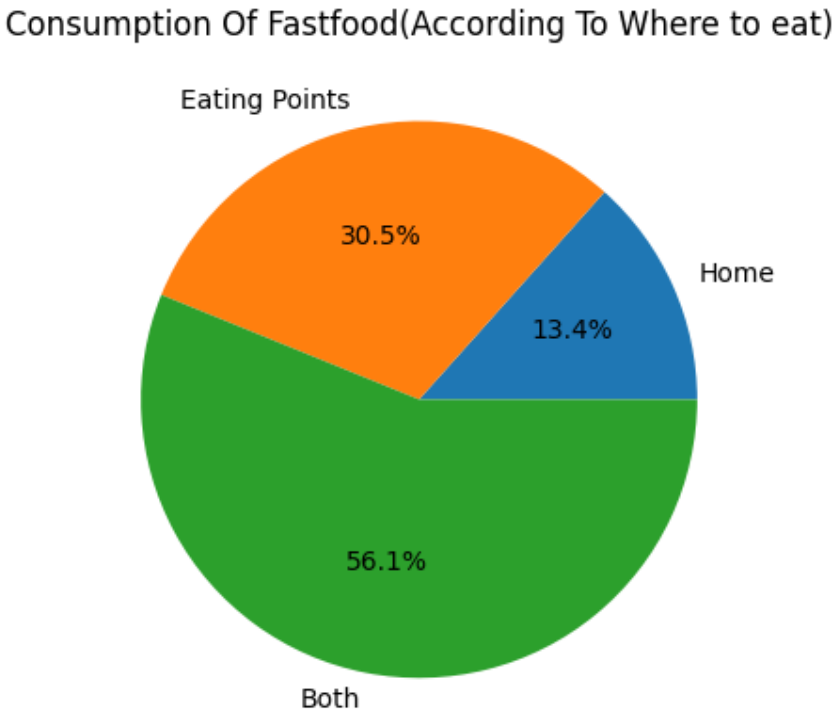
	FREQUENCY	PERCENTAGE
FAMILY	72	28.8
FRIENDS	171	63.6
ALONE	26	9.7
TOTAL	269	100



INTERPRETATION: Proportion of people consuming fast-food with friends (i.e.,63.6%) is more than proportion of people consuming fast-food alone (i.e.,9.7%).

4]

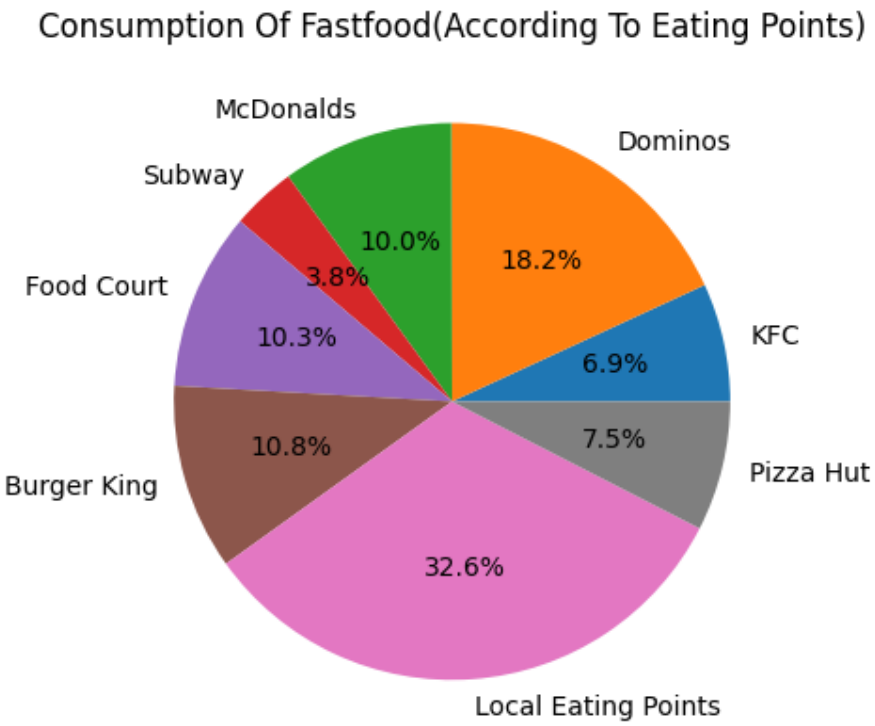
	FREQUENCY	PERCENTAGE
HOME	36	
EATING POINTS	82	
BOTH	151	
TOTAL	269	



INTERPRETATION: greater proportion of people (i.e.,56.1%) consuming fast-food at both eating points and home is greater than proportion of people consuming fast-food at home (i.e.,13.4%).

5]

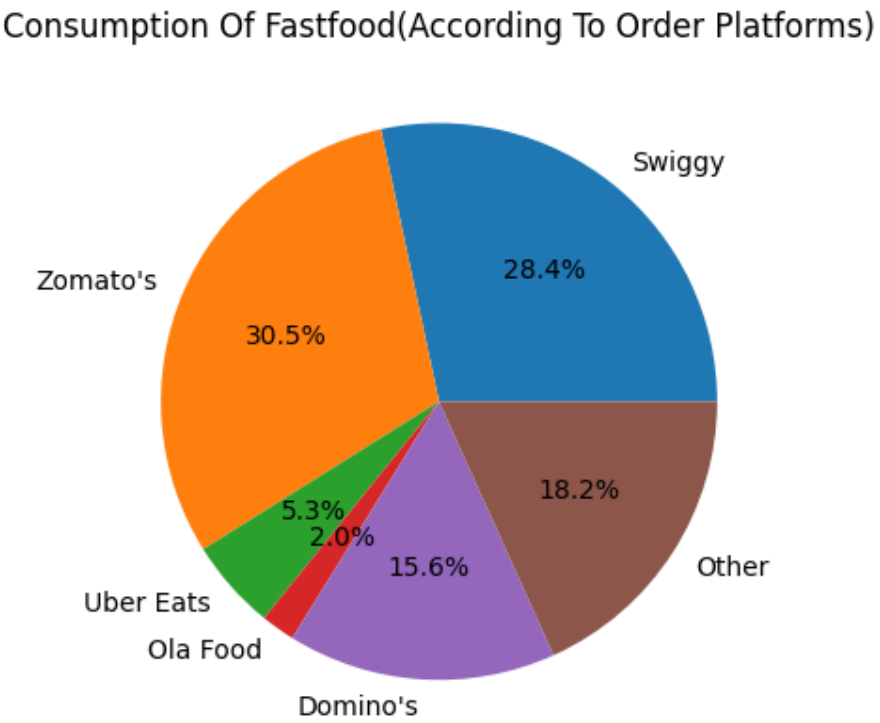
	KFC	DOMINOS	MCDONALDS	SUBWAY	FOOD COURT	BURGER KING	LOCAL EATING POINTS	PIZZA HUT	TOTAL
FREQUENCY	42	111	61	23	63	66	199	46	431
PERCENTAGE	6.9	18.2	10	3.8	10.3	10.8	32.6	7.5	100



INTERPRETATION: 32.6% of people prefer local eating points.

6]

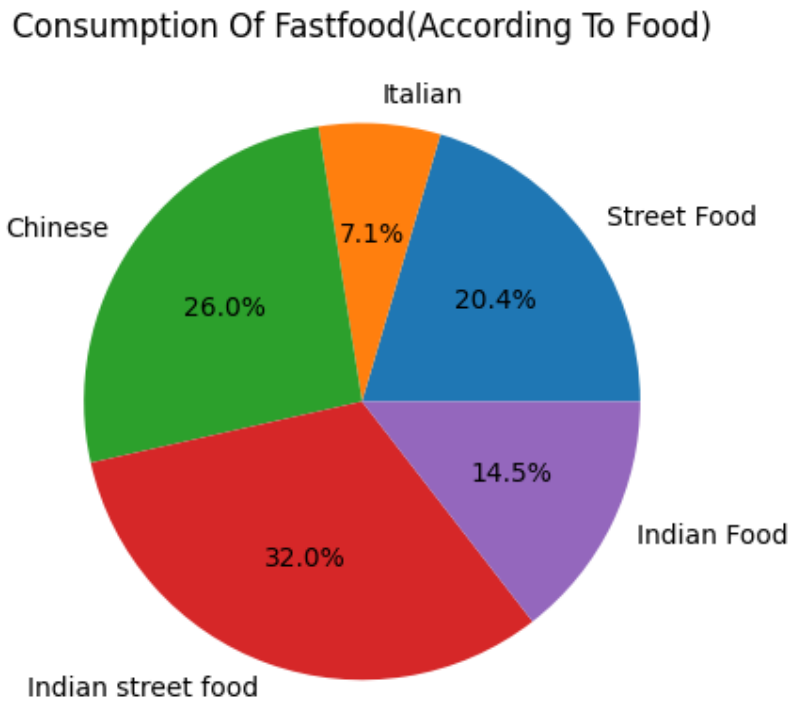
	FREQUENCY	PERCENTAGE
SWIGGY	129	28.4
ZOMATO	139	30.5
UBER EATS	24	5.3
OLA FOOD	9	2
DOMINOS	71	15.6
OTHER	83	18.2
TOTAL	455	100



INTERPRETATION: 30.5% of the people prefer Zomato for ordering fast-food.

7]

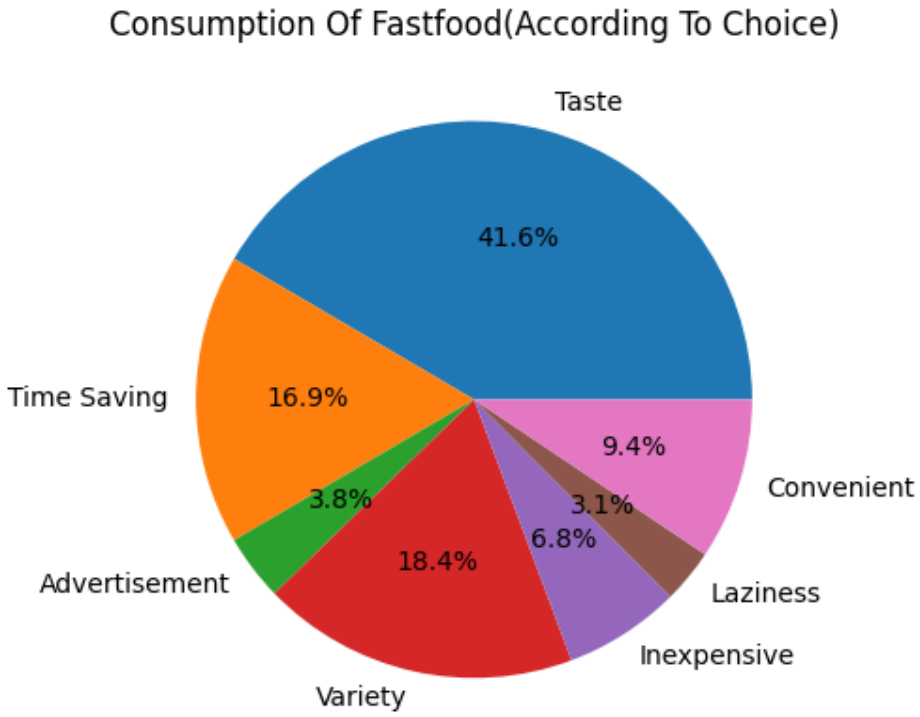
	FREQUENCY	PERCENTAGE
STREET FOOD	55	20.4
ITALIAN	19	7.1
CHINESE	70	26
INDIAN STREET FOOD	86	32
INDIAN FOOD	39	14.5
TOTAL	269	100



INTERPRETATION: 32% of the people preferred Indian Street food.

8]

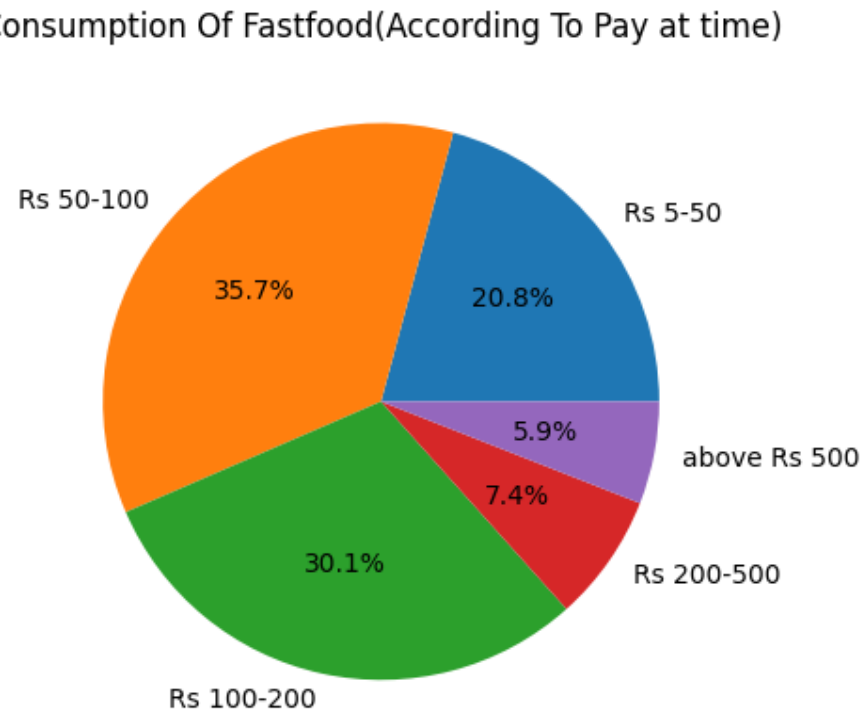
	FREQUENCY	PERCENTAGE
TASTE	239	41.6
TIME SAVINGS	97	16.9
ADVERTISEMENT	22	3.8
VARIETY	106	18.4
INEXPENSIVE	39	6.8
LAZINESS	18	3.1
CONVENIENT	54	9.4
TOTAL	575	100



INTERPRETATION: 41.6% of people consumed fast-food because of taste.

9]

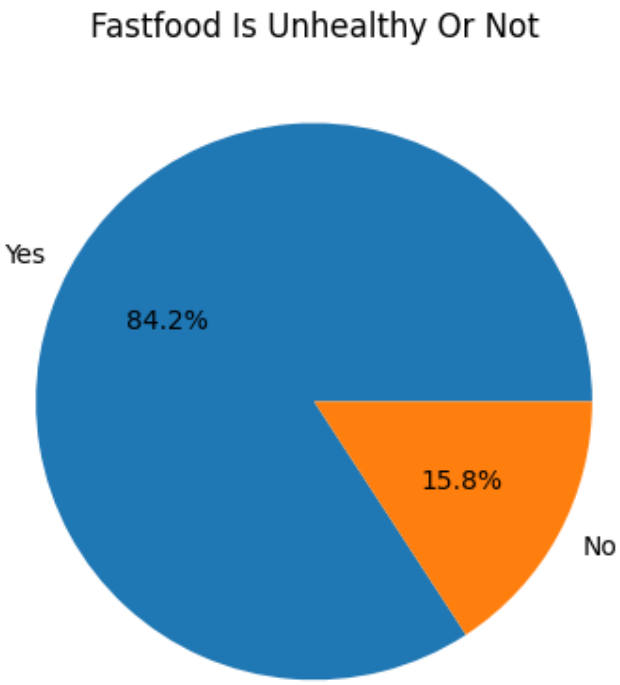
	FREQUENCY	PERCENTAGE
5-50 RS	56	20.8
50-100 RS	96	35.7
100-200 RS	81	30.1
200-500 RS	20	7.4
500 & ABOVE	16	5.9
TOTAL	269	100



INTERPRETATION: 35.7% of people are willing to spend Rs 50-100. For consumption of fast-food.

10]

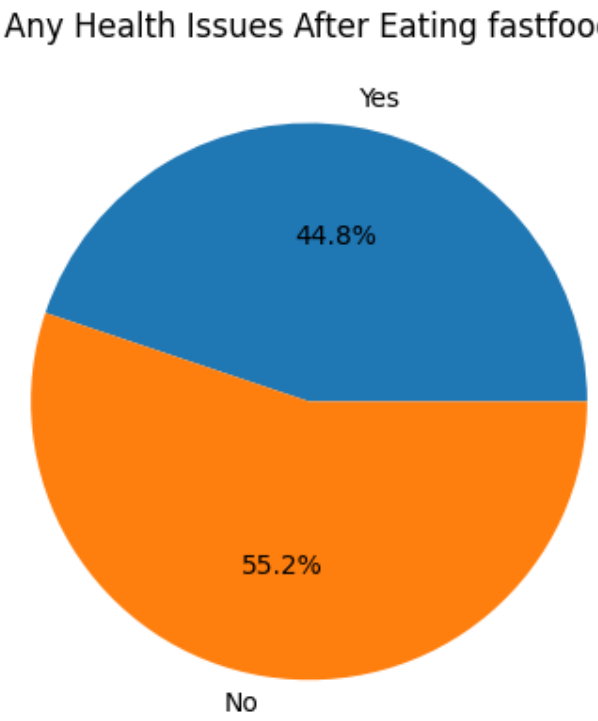
	FREQUENCY	PERCENTAGE
YES	261	84.2
NO	49	15.8
TOTAL	310	100



INTERPRETATION: 84.2% of people believe that fast-food is unhealthy.

11]

	FREQUENCY	PERCENTAGE
YES	139	44.8
NO	171	52.2
TOTAL	310	100



INTERPRETATION: 55.2% of people faced health issues after eating fast-food.

STATISTICAL ANALYSIS

Statistical analysis is the science of collecting, exploring, and presenting large amount of data to discover underlying patterns and trends. It deals with all aspects of data including the planning of data collection in terms of the design of surveys & experiments.

As this Project is Statistical Analysis on Consumption of Fast-food & Its Impact on Health. So, the analysis report will start with the testing of independence of consumption of fast-food with respect to the various categories like gender, age, residential area, residential status, occupation & Family income.

TESTING OF INDEPENDENCE OF ATTRIBUTES

❖TEST-1:

H_0: Consumption of fast food is independent of age

H_1: Consumption of fast food is dependent on age

```
X=c(19,189,53,8,0,33,6,2)
m=4
n=2
y=matrix (x, nrow = m, ncol = n)
chat=chisq.test(y)
chat
```

Decision Rule: p-value is 0.2313 which is less than l. o. s

#Reject H_0 5% l. o. s

Conclusion: consumption of fast food is dependent of age

❖TEST – 2 :

H_0: Consumption of fast food is independent of gender

H_1: Consumption of fast food is dependent on gender

```
X=c(117,152,19,22)
m=2
n=2
y=matrix (x,nrow=m,ncol=n)
char=chisq.test(y)
chat
```

Decision Rule: p-value is 0.8623which is greater than l.o.s

#Accept H_0 at 5% l.o.s

Conclusion: consumption of fast food is independent of gender

❖TEST – 3 :

H_0: Consumption of fast food is independent of residential area

H_1: Consumption of fast food is dependent of residential area

```
X=c(89,120,23,18)
m=2
n=2
y=matrix (x, nrow = m, ncol = n)
chat=chisq.test(y)
chat
```

Decision Rule: p-value is 0.1558 which is less than l. o. s

#Reject H_0 5% l. o. s

Conclusion: Consumption of fast food is dependent of residential area

❖TEST – 4:

H_0: Consumption of fast food is independent of occupation

H_1: Consumption of fast food is dependent of occupation

X=c(197,35,10,5,12,6,4,36,3,0,0,2,0,0)

m=7

n=2

y=matrix (x, nrow = m, ncol = n)

chat=chisq.test(y)

chat

Decision Rule: p-value is 0.4494 which is less than l. o. s

#Reject H_0 at 5% l. o. s

Conclusion: consumption of fast food is dependent of occupation.

❖TEST – 5:

H_0: Consumption of fast food is independent of residential status

H_1: Consumption of fast food is dependent of residential status

X=c(209,60,31,10)

m=2

n=2

y=matrix (x, nrow = m, ncol = n)

chat=chisq.test(y)

chat

Decision Rule: p-value is 0.9227 which is greater than l. o. s

#Accept H_0 at 5% l. o. s

Conclusion: Consumption of fast food is independent of residential status.

❖TEST – 6:

H_0: Consumption of fast food is independent of family income

H_1: Consumption of fast food is dependent of family income

```
X=c(59,118,37,55,18,11,2,10)
m=4
n=2
y=matrix (x, nrow = m, ncol = n)
chat=chisq.test(y)
chat
```

Decision Rule: p-value is 0.00778 which is less than l. o. s.

#Reject H_0 at 5% l. o. s.

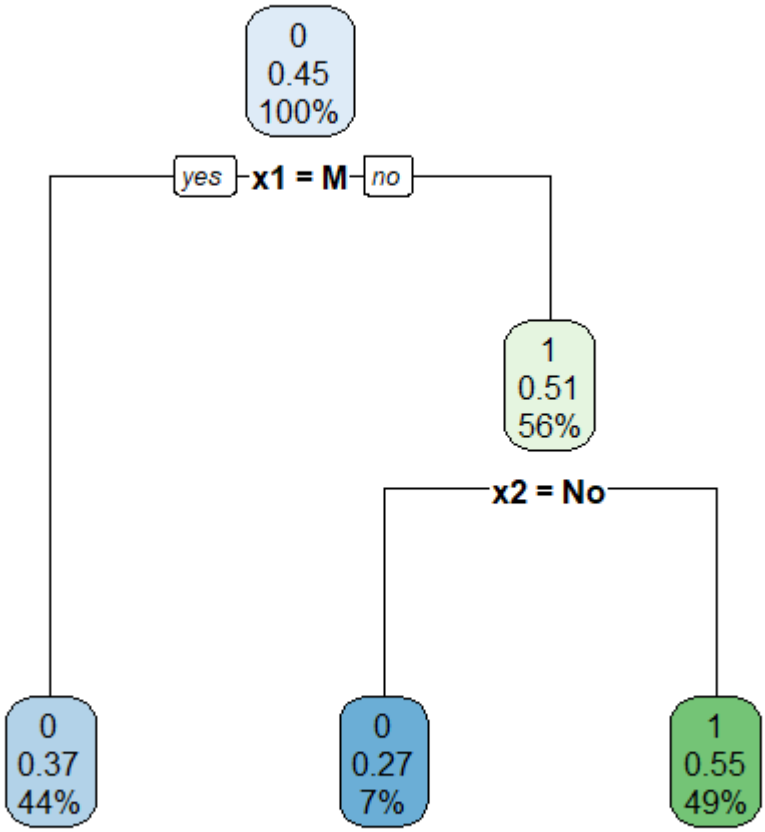
Conclusion: consumption of fast food is dependent of family income.

DECISION TREE

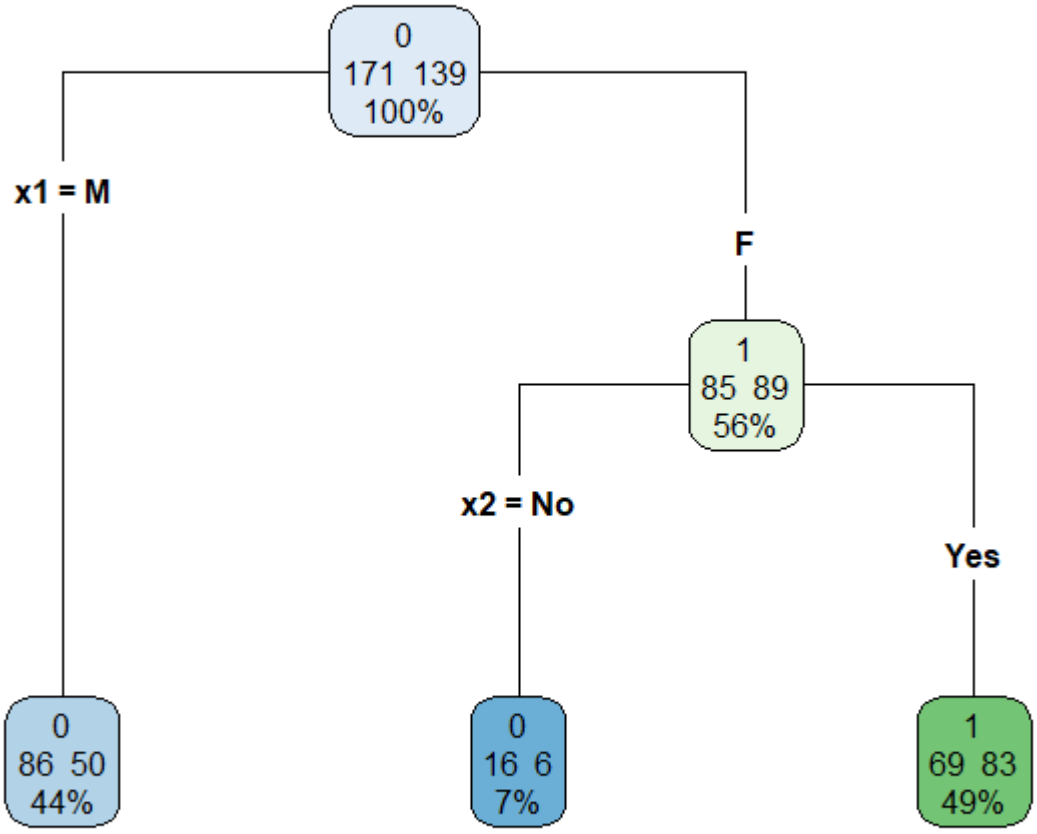
❖FITTING OF DECISION TREE ALGORITHM:

```
##FITTING OF DECISION TREE ALGORITHM
>library("readxl")
>data= read_excel ("C:\\Users\\ishwari\\Downloads\\Binary.Data.xlsx")
>data
> attach(data)
> library(rpart)
> model = rpart(y~.,data,method="class")
> model
## n= 310
##node), split, n, loss, yval, (yprob)
## * denotes terminal node
##1) root 310 139 0 (0.5516129 0.4483871)
##2) x1=M 136  50 0 (0.6323529 0.3676471) *
##3) x1=F 174  85 1 (0.4885057 0.5114943)
##6) x2=No 22   6 0 (0.7272727 0.2727273) *
##7) x2=Yes 152  69 1 (0.4539474 0.5460526) *

##PLOTTING DECISION TREE:
> library(rpart.plot)
> rpart.plot(model)
```



> rpart.plot(model,type=4,extra=101)



PREDICTION OF Y FOR X1 = M & X2 = YES

> pred_cl = predict(model, data, type="class")

> pred_cl

1

0

Levels: 0 1

INTERPRETATION = Out of 310 people proportion of male consuming fast food is 0.45 and proportion of female is 0.51 out of which 7% female face health issues after consuming fast food and remaining 49% do not face any type of health issues.

CONCLUSIONS

The survey was conducted on a sample of 310 people.

After applying the suitable tests on the data and analysing it we observed the following results:

1) Consumption Of Fast-Food is independent on Gender, age, Residential area, Residential status and family income.
2) Out of the various reasons for the consumption Of Fast-Food like Taste, Advertisement, Variety, Inexpensive, Time Saving, Convenient and Laziness, "**Taste**" is the most preferred reason among the people to consume Fast-Food.
3) From the various Fast-Food listed in our schedule like Chinese, Italian, Street Food, Indian Street Food, Indian Food, "**Chinese**" is the most preferred Fast-Food among the people.
4) Out of **310** people **82** prefer to eat Fast-Food at Eating points, **151** people like to eat at Eating points as well as Home.
5) Out of the various Eating Points like McDonald's, Domino's, Pizza Hut, Subway, KFC, Food Court and "**Local Eating Points**" are the most popular among the people to consume Fast-Food and when it comes to the preferred time for eating Fast-Food is **Evening time**.
6) The average amount of money that people spend on Fast-Food is nearly **50-100 Rs** in one time.
7) Out of **310** people **261** said that consuming Fast-Food is unhealthy and **139** people have faced health issues.


Health
is
Wealth

SCOPE AND LIMITATIONS

SCOPE:

1] Future study can be done by considering the population of India.

2] Further, research can be done to study the role of mood modification on having selfitis.

3] The follow up study can be useful to access the value of awareness program on prevention of excess selfie-taking behaviour especially for children & youth.

4] Other variables can also be considered for future study such as depression, inferiority, complex exploratory.

LIMITATIONS:

Every study has limitations. The sample was a self-selecting convenience sample of Nashik city. Our sample do not represent the population. Lack of resources was also a major concerned with the time bound. There are tools which can better analyse these data but that is out of our syllabus.

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3] Python Basics by H. Bhasin.

4] Data Analytics & Visualization Using Python by Ossama Embarak.

5] https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4484093/#:~:text=Consumption%20of%20fast%20foods%20has,annual%20total%20spending%20%5B1%5D

APPENDIX

STATISTICAL ANALYSIS OF CONSUMPTION OF FAST-FOOD & IT's IMPACT ON HEALTH

PERSONAL INFORMATION: -

Q: 1] Age:

- Below 10
- 10 – 25
- 25 – 50
- Above 50

Q: 2] Gender:

- Male
- Female

Q: 3] Residential Area:

- Rural Area
- Urban Area

Q: 4] Residential Status:

- Home
- PG/Hostel/Room

Q: 5] Occupation:

- Student
- Private sector employee
- Govt. Sector employee
- Unemployed
- Housewife
- Business
- Professor

Q: 6] Family Income:

- Below 50,000
- 50,000 - 1 lakh
- 1 lakh – 2 lakhs
- Above 2 lakhs

Q: 7] Do you consume Fast-food?

- Yes
- No

Q: 8] How often do you consume Fast-food on weekly basis? (If yes)

- 1 - 2 times
- 3 – 4 times
- 5 – 6 times
- More than 7 times

Q: 9] What time do you like to eat fast-food? (If yes)

- Morning (Breakfast)
- Afternoon (Lunch)
- Evening (snack)
- Night (Dinner)

Q: 10] With whom you like to visit fast-food corner? (If yes)

- Family
- Friends
- Alone

Q: 11] Where do you prefer to eat fast-food? (If yes)

- Home
- Eating points
- Both

Q: 12] Which eating points do you prefer? (If yes)

- KFC
- Domino's
- McDonald's
- Subway
- Food court
- Pizza Hut
- Burger King
- Local eating points

Q: 13] Which platform do you prefer to order fast-food? (If yes)

- Swiggy
- Zomato
- UberEATS
- Ola food

- Domino’s
- Other

Q: 14] Which food do you prefer? (If yes)

- Street food
- Italian
- Chinese
- Indian Street food
- Indian food

Q: 15] What makes you like to eat fast-food? (If yes)

- Taste
- Inexpensive
- Advertisement
- Variety
- Time saving
- Laziness
- Convenient

Q: 16] On an average how much would you expect to pay for a fast-food meal? (If yes)

- 5 – 50 Rs
- 50 – 100 Rs
- 100 – 200 Rs
- 200 – 500 Rs
- Above 500 Rs

Q: 17] Is Fast-food unhealthy?

- Yes
- No

Q: 18] Do you face any health issues after eating fast-food?

- Yes
- No